EXERCISES COURSE

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1. Week 2 (3.1)

Problem 1.1. *Calculate* $\displaystyle\int_0^2 \frac{\mathrm{d}x}{\sqrt{(4-x^2)(9-x^2)}}$.

Solution. Let $x = 2\sin\theta$. The integral becomes

$$\int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{9-4\sin^2\theta}} = \frac{1}{3}\int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{1-\frac{4}{9}\sin^2\theta}} = \frac{1}{3}F\left(\frac{2}{3}, \frac{\pi}{2}\right).$$

□

Problem 1.2. *Calculate* $\displaystyle\int_0^1 \frac{\mathrm{d}x}{\sqrt{(1+x^2)(1+2x^2)}}$.

Solution. Let $x = \tan\theta$. The integral becomes

$$\begin{aligned}\int_0^{\pi/4} \frac{\sec^2\theta \mathrm{d}\theta}{\sqrt{1+\tan^2\theta}\sqrt{1+2\tan^2\theta}} &= \int_0^{\pi/4} \frac{\mathrm{d}\theta}{\sqrt{\cos^2\theta + 2\sin^2\theta}} \\ &= \int_0^{\pi/4} \frac{\mathrm{d}\theta}{\sqrt{2-\cos^2\theta}} \\ &= \frac{1}{\sqrt{2}}\int_0^{\pi/4} \frac{\mathrm{d}\theta}{\sqrt{1-\frac{1}{2}\cos^2\theta}}.\end{aligned}$$

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Let $\phi = \pi/2 - \theta$. The integral becomes

$$\frac{1}{\sqrt{2}} \int_{\pi/4}^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{1 - \frac{1}{2}\sin^2\phi}} = \frac{1}{\sqrt{2}} \left\{ F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{2}\right) - F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right) \right\}.$$

□

Problem 1.3. *Calculate* $\displaystyle\int_4^6 \frac{\mathrm{d}x}{\sqrt{(x-1)(x-2)(x-3)}}$.

Solution. Let $u = \sqrt{x-3}$ or $x = 3 + u^2$. The integral becomes

$$2\int_1^{\sqrt{3}} \frac{\mathrm{d}u}{\sqrt{(u^2+2)\,(u^2+1)}}.$$

Let $u = \tan\theta$. The integral becomes

$$\begin{aligned} 2\int_{\pi/4}^{\pi/3} \frac{\mathrm{d}\theta}{\sqrt{2\cos^2\theta + \sin^2\theta}} &= 2\int_{\pi/4}^{\pi/3} \frac{\mathrm{d}\theta}{\sqrt{2 - \sin^2\theta}} = \sqrt{2}\int_{\pi/4}^{\pi/3} \frac{\mathrm{d}\theta}{\sqrt{1 - \frac{1}{2}\sin^2\theta}} \\ &= \sqrt{2}\left\{ F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{3}\right) - F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right) \right\}. \end{aligned}$$

□

Problem 1.4. *Calculate* $\displaystyle\int \sqrt{\sin x}\,\mathrm{d}x$.

Solution.

$$\begin{aligned} \int \sqrt{\sin x}\,\mathrm{d}x &= \int \sqrt{\cos\left(x - \frac{\pi}{2}\right)}\,\mathrm{d}x \\ &= \int \sqrt{1 - 2\sin^2\left(\frac{x}{2} - \frac{\pi}{4}\right)}\,\mathrm{d}x \\ &= 2\int \sqrt{1 - 2\sin^2\varphi}\,\mathrm{d}\varphi \\ &= 2E(\sqrt{2}, \varphi). \end{aligned}$$

□

Problem 1.5. *Calculate* $\displaystyle\int_0^{2\pi} \sqrt{2 + \sin 2\theta}\,\mathrm{d}\theta$.

Solution.

$$\begin{aligned} \int_0^{2\pi} \sqrt{2 + \sin 2\theta}\,\mathrm{d}\theta &= 2\int_0^{\pi} \sqrt{2 + \sin 2\theta}\,\mathrm{d}\theta \\ &= 2\int_0^{\pi} \sqrt{2 + \cos 2\left(\theta - \frac{\pi}{4}\right)}\,\mathrm{d}\theta \end{aligned}$$

$$
\begin{aligned}
&= 2\int_0^{2\pi} \sqrt{3 - 2\sin^2\left(\theta - \frac{\pi}{4}\right)}\,\mathrm{d}\theta \\
&= 2\sqrt{3}\int_{-\frac{\pi}{4}}^{\frac{3}{4}\pi} \sqrt{1 - \frac{2}{3}\sin^2\theta}\,\mathrm{d}\theta \\
&= 2\sqrt{3}\int_0^{\frac{\pi}{4}} \sqrt{1 - \frac{2}{3}\sin^2\theta}\,\mathrm{d}\theta + 2\sqrt{3}\int_0^{\frac{3}{4}\pi} \sqrt{1 - \frac{2}{3}\sin^2\theta}\,\mathrm{d}\theta \\
&= 2\sqrt{3}\left(E\left(\frac{\sqrt{6}}{3}, \frac{\pi}{4}\right) + E\left(\frac{\sqrt{6}}{3}, \frac{3\pi}{4}\right)\right).
\end{aligned}
$$

□

Problem 1.6. *Calculate the perimeter of an ellipse.*

Solution. The ellipse $x = a\cos\theta$, $4y = b\sin\theta$, $a > b > 0$, has length

$$
\begin{aligned}
L &= 4\int_0^{\pi/2} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} = 4\int_0^{\pi/2} \sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\mathrm{d}\theta \\
&= 4\int_0^{\pi/2} \sqrt{a^2 - (a^2 - b^2)\sin^2\theta}\mathrm{d}\theta = 4a\int_0^{\pi/2} \sqrt{1 - e^2\sin^2\theta}\mathrm{d}\theta,
\end{aligned}
$$

where $e^2 = (a^2 - b^2)/a^2 = c^2/a^2$ is the square of the eccentricity of the ellipse. The result can be written as

$$
L = 4aE\left(e, \frac{\pi}{2}\right) = 4aE(e).
$$

For the special case of a circle, $a = b = r$, i.e., $e = 0$, and $E(0) = \pi/2$, and we recover the circumference of a circle: $L = 2\pi r$. □

Remark 1.7. *The term elliptic integral was coined by Count Fagnano (1682-1766) in 1750. He discovered that the arclength of the lemniscate can be expressed in terms of an elliptic integral of the first kind.*

Problem 1.8. *Calculate the arclength of a lemniscate.*

Solution. The lemniscate is the curve:

$$
\left(x^2 + y^2\right)^2 = a^2\left(x^2 - y^2\right)^2,
$$

or in polar form

$$
r^2 = a^2\cos 2\theta.
$$

From $\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2$,

$$
\begin{aligned}
L = 4\int_{r=0}^{\pi/4} \mathrm{d}s &= 4a\int_0^{\pi/4} \frac{\mathrm{d}\theta}{\sqrt{\cos 2\theta}}, \quad \left(\cos 2\theta = \cos^2 u\right) \\
&= \int_0^{\pi/2} \frac{\mathrm{d}u}{\sqrt{2 - \sin^2 u}} = \frac{1}{\sqrt{2}}\int_0^{\pi/2} \frac{\mathrm{d}u}{\sqrt{1 - \frac{1}{2}\sin^2 u}} = \frac{1}{\sqrt{2}} \cdot F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{2}\right).
\end{aligned}
$$

Thus,

$$L = 4a \cdot \frac{1}{\sqrt{2}} F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{2}\right) = a \cdot 2\sqrt{2}(1.85407) = 5.244102a.$$

□

Remark 1.9. *(Historical note: The rectification of the lemniscate was first done by Fagnano in 1718. The lemniscatus, L. 'decorated by ribbons', was first studied in astronomy in 1680 by Cassini, known as the ovals of Cassini, but his book was published in 1749, many years after his death. The curves were popularized by the Bernoulli brothers in 1694.) Cassini considered more general forms of the lemniscate for whose points the products of the distances to two foci is a constant:*

$$d_1 d_2 = b^2,$$

$$b^4 = r^4 + \frac{a^2}{4} - r^2 a^2 \cos 2\theta.$$

When $b = \frac{a}{\sqrt{2}}$ centered at the origin, we get the ribbon-shaped curve.

Exercise 1.10 (Leave to the reader)**.** *Calculate the finite-amplitude pendulum*

Solution. The equation of motion is:

$$ml\ddot{\theta} = -mg\sin\theta.$$

Let $p = \dot{\theta}$, we have

$$p\frac{\mathrm{d}p}{\mathrm{d}\theta} = -\frac{g}{l}\sin\theta,$$

which yields

$$\frac{p^2}{2} = \frac{g}{l}\cos\theta + C.$$

For the initial condition, $t = 0 : \theta = \theta_0$, $\dot{\theta} = 0$, we have

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\sqrt{\frac{2g}{l}}\sqrt{\cos\theta - \cos\theta_0}.$$

The period, T, is given by

$$\frac{T}{4} = -\sqrt{\frac{l}{2g}}\int_{\theta_0}^{0} \frac{\mathrm{d}\theta}{\sqrt{\cos\theta - \cos\theta_0}}$$

$$\text{or,} \quad T = 4\sqrt{\frac{l}{2g}}\int_{0}^{\theta_0} \frac{\mathrm{d}\theta}{\sqrt{\cos\theta - \cos\theta_0}} = 2\sqrt{\frac{l}{g}}\int_{0}^{\theta_0} \frac{\mathrm{d}\theta}{\sqrt{\sin^2(\theta_0/2) - \sin^2(\theta/2)}}$$

$$= 4\sqrt{\frac{l}{g}}\int_{0}^{\pi/2} \frac{\mathrm{d}u}{\sqrt{1 - k^2\sin^2 u}}, \ \sin\left(\frac{\theta}{2}\right) = \sin\frac{\theta_0}{2} \cdot \sin u, \ k = \sin\left(\frac{\theta_0}{2}\right).$$

Hence

$$T = 4\sqrt{\frac{l}{g}} \cdot F\left(k, \frac{\pi}{2}\right),$$

an elliptic integral. For the special case of smalloscillations, $k = 0$, we get the classical result:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$\square$

2. Week 3 (3.8)

Problem 2.1. *Calculate the area of regions closed by the following curves.*

(1) *The cycloid:* $x = a(t - \sin t)$, $y = a(1 - \cos t)$ $(0 \le t \le 2\pi,\ a > 0)$ *and* x*-axis.*
(2) *The evolvent of sphere* $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$ $(0 \le t \le 2\pi,\ a > 0)$ *and* $x = a$.
(3) *The rhodonea curve* $r = a\sin 3\theta$.
(4) *The folium of Descartes* $r = \dfrac{3a\sin\theta\cos\theta}{\sin^3\theta + \cos^3\theta}$.

Solution. (1)

$$\begin{aligned} S = \int_0^{2\pi a} y\,\mathrm{d}x &= \int_0^{2\pi} a(1-\cos t)\cdot a(1-\cos t)\,\mathrm{d}\theta \\ &= a^2\int_0^{2\pi}(\cos^2 t - 2\cos t + 1)\,\mathrm{d}\theta \\ &= 3\pi a^2. \end{aligned}$$

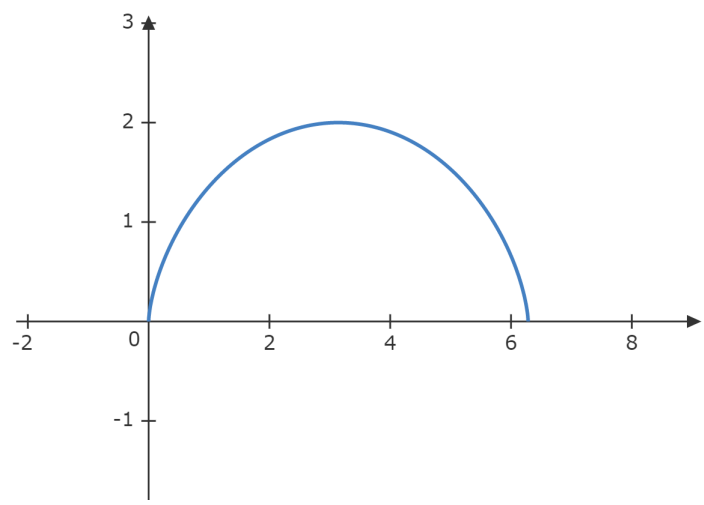

Figure 1. Graph of the cycloid for $a = 1$

(2)

$$\begin{aligned} S = \int (x-a)\,\mathrm{d}y &= a^2\int_0^{2\pi}(\cos t + t\sin t - 1)(\cos t - \cos t + t\sin t)\,\mathrm{d}t \\ &= a^2\int_0^{2\pi}\left(t\sin t\cos t + t^2\sin^2 t - t\sin t\right)\,\mathrm{d}t \\ &= \frac{a^2}{3}(4\pi^3 + 3\pi) \end{aligned}$$

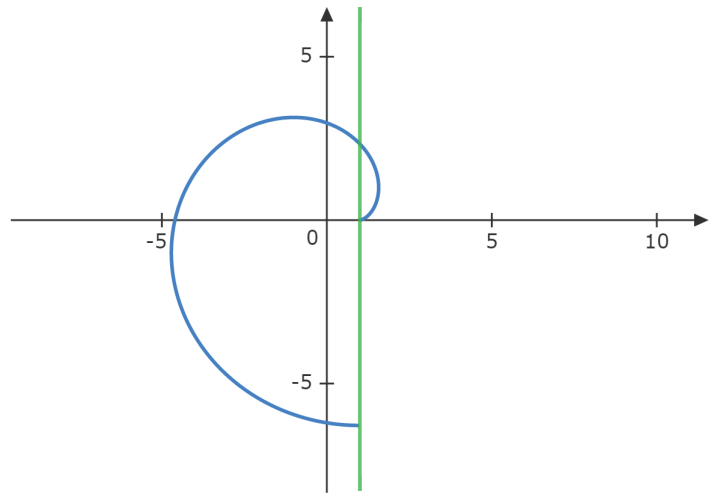

FIGURE 2. Graph of the evolvent for $a = 1$

(3)

$$\begin{aligned} S &= 3 \cdot \frac{1}{2} \int_0^{\frac{\pi}{3}} a^2 \sin^2 3\theta \, \mathrm{d}\theta \\ &= 3 \cdot \frac{a^2}{2} \int_0^{\frac{\pi}{3}} \frac{1 - \cos 6\theta}{2} \, \mathrm{d}\theta \\ &= \frac{\pi}{4} a^2. \end{aligned}$$

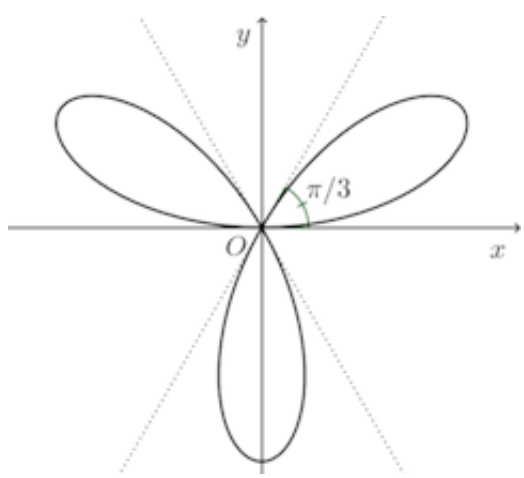

FIGURE 3. Graph of the rhodonea curve for $a = 1$

(4)

$$
\begin{aligned}
S &= \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{9a^2 \sin^2 \cos^2 \theta}{\left(\sin^3\theta + \cos^3\theta\right)^2}\,\mathrm{d}\theta \\
&= \frac{9a^2}{2}\int_0^{\frac{\pi}{2}} \frac{\tan^2\theta}{\left(\tan^3\theta + 1\right)^2}\,\mathrm{d}\tan\theta \\
&= \frac{3a^2}{2}\int_0^{\frac{\pi}{2}} \frac{1}{\left(\tan^3\theta + 1\right)^2}\,\mathrm{d}\tan\theta \\
&= \frac{3a^2}{2}\frac{-1}{\tan^3\theta + 1}\bigg|_0^{\frac{\pi}{2}} = \frac{3a^2}{2}.
\end{aligned}
$$

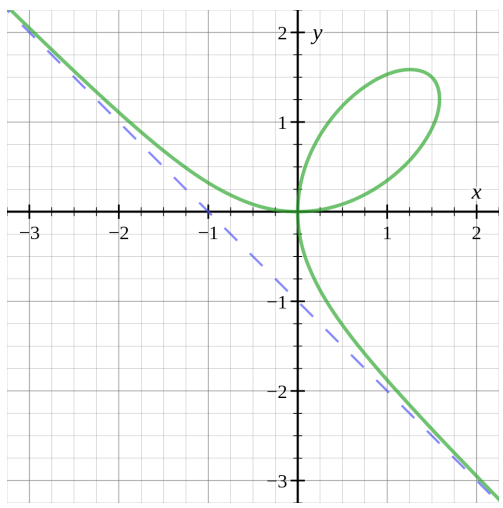

FIGURE 4. Graph of the folium of Descartes for $a = 1$

□

Problem 2.2 (7.43)**.** *Find volumes of the solid of revolution obtained by rotating the region closed by $y = x^2$ $(0 \le x \le h)$, $y = 0$ and $x = h$ around x-axis, y-axis, respectively.*

Solution. Around x-axis:

$$V = \pi\int_0^h y^2\,\mathrm{d}x = \pi\int_0^h x^4\,\mathrm{d}x = \frac{\pi}{5}h^5.$$

Around y-axis: **First way**:

$$V = \pi\int_0^h (h^2 - x^2)\,\mathrm{d}y = \pi\int_0^h (h^2 - x^2)2x\,\mathrm{d}x = \frac{\pi}{2}h^4.$$

Second way:

$$V = 2\pi\int_0^h xy\,\mathrm{d}x = 2\pi\int_0^h x^3\,\mathrm{d}x = \frac{\pi}{2}h^4.$$

□

Problem 2.3 (7.49)**.** *Calculate the arc length of the following curves.*

(2) *Archimedean spiral:* $r = a\theta$ $(0 \le \theta \le 2\pi)$.

(4) *The evolvent of sphere* $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$ $(0 \le t \le 2\pi,\ a > 0)$.

Solution. (2)

$$\begin{aligned} L = \int_0^{2\pi} \sqrt{r^2 + r'^2}\,\mathrm{d}\theta &= a\int_0^{2\pi} \sqrt{1+\theta^2}\,\mathrm{d}\theta \\ &= \frac{a}{2}\theta\sqrt{1+\theta^2} + \frac{a}{2}\ln(\theta + \sqrt{1+\theta^2})\Big|_0^{2\pi} \\ &= a\pi\sqrt{1+4\pi^2} + \frac{a}{2}\ln(4\pi + \sqrt{1+4\pi^2}). \end{aligned}$$

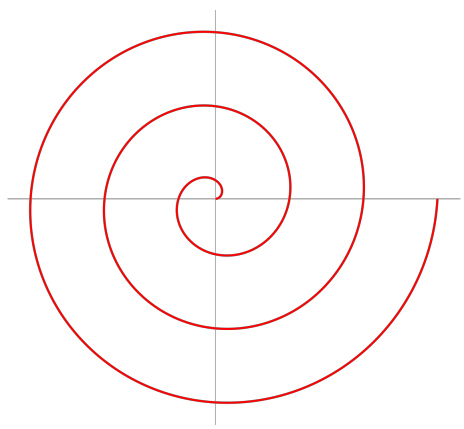

FIGURE 5. Graph of the Archimedean spiral for $a = 1$

(4)

$$\begin{aligned} L = \int_0^{2\pi} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} &= a\int_0^{2\pi} \sqrt{(t\cos t)^2 + (t\sin t)^2}\,\mathrm{d}t \\ &= a\int_0^{2\pi} t\,\mathrm{d}t = 2\pi^2 a. \end{aligned}$$

□

Problem 2.4 (7.51(4))**.** *Calculate the area of the surface of revolution obtained by rotating the curve* $y = \sin x\,(0 \le x \le \pi)$ *around* x*-axis.*

Solution.

$$\begin{aligned} dS &= 2\pi y\sqrt{y'^2 + 1}\,\mathrm{d}x \\ &= 2\pi\sin x\sqrt{1+\cos^2 x}\,\mathrm{d}x. \end{aligned}$$

$$
\begin{aligned}
S &= 2\pi \int_0^{\pi} -\sqrt{1+\cos^2 x}\,\mathrm{d}\cos x \\
&= 4\pi \int_0^1 \sqrt{1+t^2}\,\mathrm{d}t = 2\sqrt{2}\pi + 2\pi \ln(\sqrt{2}+1).
\end{aligned}
$$

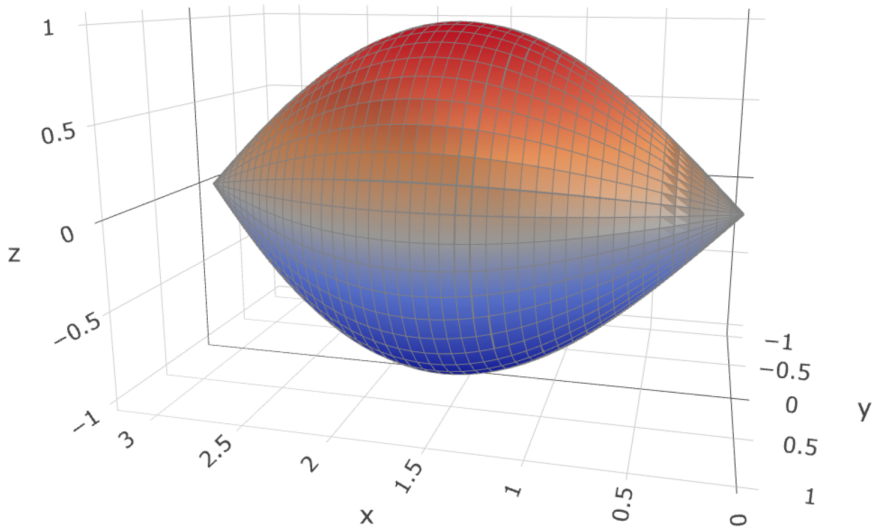

FIGURE 6. $y = \sin x$ rotated around x-axis

□

Problem 2.5. *Find the area of the region determined by* $x^2 + xy + y^2 = 1$.

Solution. **First way:** From the equation, we have

$$
y_{1,2}(x) = -\frac{x}{2} \pm \sqrt{1 - \frac{3}{4}x^2}, \quad -\frac{2}{\sqrt{3}} \le x \le \frac{2}{\sqrt{3}}.
$$

Then

$$
\begin{aligned}
S &= \int_{-2/\sqrt{3}}^{2/\sqrt{3}} [y_1(x) - y_2(x)]\,\mathrm{d}x = 2\int_{-2/\sqrt{3}}^{2/\sqrt{3}} \sqrt{1 - \frac{3}{4}x^2}\,\mathrm{d}x \\
&= \frac{4}{\sqrt{3}} \int_{-\pi/2}^{\pi/2} \mathrm{d}\theta = \frac{2\pi}{\sqrt{3}}.
\end{aligned}
$$

Second way: Let $x = r\cos\theta$, $y = r\sin\theta$, we have

$$
r^2 = \frac{1}{1 + \sin\theta\cos\theta}.
$$

Then

$$
S = \frac{1}{2}\int_0^{2\pi} r^2\,\mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi} \frac{\mathrm{d}\theta}{1+\sin\theta\cos\theta} = \int_0^{2\pi} \frac{\mathrm{d}\theta}{2+\sin\theta} = \frac{2\pi}{\sqrt{3}}.
$$

Third way: Note that

$$1 = x^2 + xy + y^2 = \frac{3}{4}x^2 + \left(y + \frac{x}{2}\right)^2.$$

Let

$$x = \frac{2}{\sqrt{3}}\cos t,\ y = \sin t - \frac{1}{\sqrt{3}}\cos t, \quad 0 \le t \le 2\pi.$$

Since

$$\begin{aligned}
&x(t)y'(t) - y(t)x'(t) \\
&= \frac{2}{\sqrt{3}}\cos t\left(\cos t + \frac{1}{\sqrt{3}}\sin t\right) - \left(\sin t - \frac{1}{\sqrt{3}}\cos t\right)\left(-\frac{2}{\sqrt{3}}\sin t\right) \\
&= \frac{2}{\sqrt{3}},
\end{aligned}$$

we have

$$S = \frac{1}{2}\int_0^{2\pi}(x\,\mathrm{d}y - y\,\mathrm{d}x) = \frac{1}{2}\int_0^{2\pi}\frac{2}{\sqrt{3}}\,\mathrm{d}t = \frac{2\pi}{\sqrt{3}}.$$

□

Problem 2.6. *Suppose that a curve is given by* $y = \int_0^x \sqrt{\sin t}\,\mathrm{d}t$, $0 \le x \le \pi$. *Calculate its arc length.*

Solution. By the formula of arc length calculation, we have

$$\begin{aligned}
L &= \int_0^\pi \sqrt{1 + f'^2(x)}\,\mathrm{d}x = \int_0^\pi \sqrt{1 + \sin x}\,\mathrm{d}x \\
&= \int_0^\pi \left(\sin\frac{x}{2} + \cos\frac{x}{2}\right)\mathrm{d}x \\
&= 2\left(\int_0^{\pi/2}\sin t\,\mathrm{d}t + \int_0^{\pi/2}\cos t\,\mathrm{d}t\right) \\
&= 4\int_0^{\pi/2}\sin t\,\mathrm{d}t = 4.
\end{aligned}$$

□

Problem 2.7. *Use Young's inequality to prove when* $a, b \ge 1$, *there is* $ab \le e^{a-1} + b\ln b$.

Proof. Choosing $f(x) = e^x - 1$, $g(y) = \ln(y+1)$, we have

$$\begin{aligned}
(a-1)(b-1) &\le \int_0^{a-1}(e^x - 1)\,\mathrm{d}x + \int_0^{b-1}\ln(y+1)\,\mathrm{d}y \\
&= e^{a-1} - (a-1) - 1 + b\ln b - (b-1) \\
&= e^{a-1} + b\ln b - a - b + 1,
\end{aligned}$$

i.e.

$$ab \le e^{a-1} + b \ln b.$$

□

Exercise 2.8 (Leave to the reader)**.** *Use Minkowski's inequality to show that*

$$\int_0^\pi |f(x) - \sin x|^2 \, \mathrm{d}x \le \frac{3}{4} \quad \textit{and} \quad \int_0^\pi |f(x) - \cos x|^2 \, \mathrm{d}x \le \frac{3}{4}$$

cannot be simultaneously true, where $f \in R[a, b]$.

Hint:

$$\begin{aligned}
\sqrt{\pi} &= \left(\int_0^\pi |\sin x - \cos x|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} \\
&= \left(\int_0^\pi |(f(x) - \cos x) - (f(x) - \sin x)|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} \\
&\le \left(\int_0^\pi |f(x) - \sin x|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} + \left(\int_0^\pi |f(x) - \cos x|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} \\
&\le \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3},
\end{aligned}$$

contradiction. □

Problem 2.9. *Find the minimum of the area enclosed by the parabola* $y^2 = 2x$ *and its chord (passing through the focus).*

Solution. Since the focus of parabola $y^2 = 2x$ is $(\frac{1}{2}, 0)$, we can assume that the equation of a chord is $x = ky + \frac{1}{2}$. Then

$$\begin{aligned}
S &= \int_{y_1}^{y_2} \left[\left(ky + \frac{1}{2} \right) - \frac{y^2}{2} \right] \mathrm{d}y \\
&= \frac{k}{2} (y_2 - y_1) \cdot (y_2 + y_1) + \frac{1}{2} (y_2 - y_1) - \frac{1}{6} (y_2 - y_1) \left(y_2^2 + y_1 y_2 + y_1^2 \right),
\end{aligned}$$

where y_1, y_2 are the intersection points of the parabola and its chord. By $y^2 = 2x$, we have

$$y^2 = 2ky + 1.$$

Hence, the Vieta theorem gives us that

$$\begin{aligned}
y_1 + y_2 &= 2k, \\
y_1 \cdot y_2 &= -1.
\end{aligned}$$

Then

$$
\begin{aligned}
S &= \frac{1}{6}\left(y_2 - y_1\right) \cdot \left[3k\left(y_2 + y_1\right) + 3 - \left(y_2^2 + y_1 y_2 + y_1^2\right)\right] \\
&= \frac{1}{6} \cdot 2 \cdot \sqrt{k^2+1} \cdot \left[3k \cdot 2k + 3 - \left(4k^2 + 1\right)\right] \\
&= \frac{1}{3}\sqrt{k^2+1} \cdot \left(2k^2 + 2\right) \\
&= \frac{2}{3} \cdot \left(k^2 + 1\right)^{\frac{3}{2}}.
\end{aligned}
$$

Hence

$$S_{\min} = \frac{2}{3}.$$

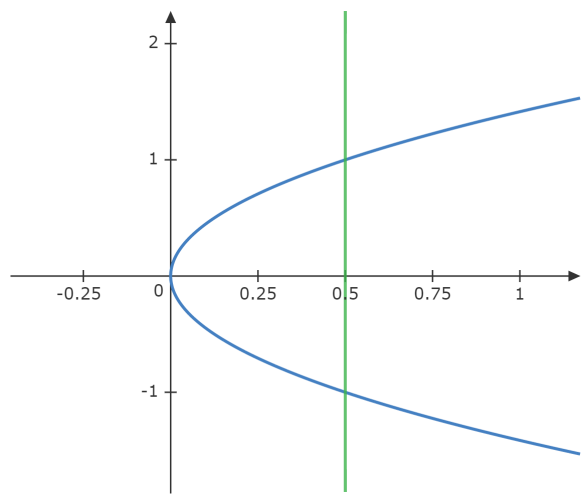

FIGURE 7. Graph of $y^2 = 2x$ and $x = \frac{1}{2}$

□

Problem 2.10 (Wirtinger's inequality)**.** *Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous differentiable function with period 2π, and satisfies $\int_0^{2\pi} f(x)\,\mathrm{d}x = 0$. Prove that*

$$\int_0^{2\pi} f'(x)^2\,\mathrm{d}x \geq \int_0^{2\pi} f(x)^2\,\mathrm{d}x.$$

"=" holds iff $f(x) = a\cos x + b\sin x$.

Proof. Firstly, we show there is a $t_0 \in [0, \pi)$ such that $f(t_0) = f(t_0 + \pi)$. Indeed, if $f(0) = f(\pi)$, we can take $t_0 = 0$. If $f(0) \neq f(\pi)$, let $g(x) = f(x) - f(x + \pi)$, then $g(0)g(\pi) < 0$, hence by the intermediate value theorem, we know that there exists a $t_0 \in [0, \pi)$ such that $g(t_0) = 0$, i.e. $f(t_0) = f(t_0 + \pi)$.

Next, let $c = f(t_0) = f(t_0 + \pi)$. It's easy to calculate that (Leave to the reader!)

$$f'^2 - (f-c)^2 - (f' - (f-c)\cot(t-t_0))^2 = \left((f-c)^2 \cot(t-t_0)\right)'.$$

Note that $(f-c)^2 \cot(t-t_0)$ is continuous at t_0 and $t_0 + \pi$, we have

$$\int_0^{2\pi} f'(t)^2 \, \mathrm{d}t - \int_0^{2\pi} (f(t)-c)^2 \, \mathrm{d}t \geq \left((f-c)^2 \cot(t-t_0)\right)\Big|_0^{2\pi} = 0.$$

So

$$\int_0^{2\pi} f'(t)^2 \, \mathrm{d}t - \int_0^{2\pi} f(t)^2 \, \mathrm{d}t \geq 2\pi c^2 \geq 0.$$

"=" holds iff $c = 0$, and $f'(t) = f(t)\cot(t-t_0)$, then $f(t) = A\sin(t-t_0)$. □

Problem 2.11 (Isoperimetric inequality)**.** *Prove the isoperimetric inequality in $\mathbb{R}^2$: $4\pi A \leq L^2$, "=" holds iff the curve is a circle.*

Proof. Suppose that $\gamma(t) = (x(t), y(t)) : [0, 2\pi] \to \mathbb{R}^2$ is a curve parameterized by arc length. Without loss of generality, we may assume that $\int_0^{2\pi} x(t)\mathrm{d}t = 0$. Then, by Problem 2.10, we have

$$\begin{aligned} L^2 - 4\pi A =& 2\pi \int_0^{2\pi} \left(x'^2 + y'^2\right) \mathrm{d}t - 4\pi \int_0^{2\pi} xy' \mathrm{d}t \\ =& 2\pi \int_0^{2\pi} \left(x'^2 - x^2 + (y' - x)^2\right) \mathrm{d}t \\ \geq& 2\pi \int_0^{2\pi} \left(x'^2 - x^2\right) \mathrm{d}t \geq 0. \end{aligned}$$

"=" holds iff $x(t) = a\cos t + b\sin t, y'(t) = x(t)$, thus it is a circle. □

Exercise 2.12. *Suppose a flat plate of uniform density has the shape contained by $y = x^2$, $y = 1$, and $x = 0$, in the first quadrant. Find the center of mass.*

Hint: We compute the moment around the y-axis:

$$M_y = \int_0^1 x\left(1 - x^2\right) dx = \frac{1}{4},$$

and the total mass

$$M = \int_0^1 \left(1 - x^2\right) dx = \frac{2}{3},$$

and finally

$$\bar{x} = \frac{1}{4} \cdot \frac{3}{2} = \frac{3}{8}.$$

Next we do the same thing to find $\bar{y}$.

$$M_x = \int_0^1 y\sqrt{y} dy = \frac{2}{5},$$

and

$$\bar{y} = \frac{2}{5} \cdot \frac{3}{2} = \frac{3}{5},$$

since the total mass M is the same. $\square$

Remark 2.13. *Since the density is constant, the center of mass depends only on the shape of the plate, not the density, or in other words, this is a purely geometric quantity. In such a case the center of mass is called the centroid.*

Exercise 2.14. *Suppose that a water tank is shaped like a right circular cone with the tip at the bottom, and has height 10 meters and radius 2 meters at the top. If the tank is full, how much work is required to pump all the water out over the top?*

Hint: Here we have a large number of atoms of water that must be lifted different distances to get to the top of the tank. Fortunately, we don't really have to deal with individual atoms–we can consider all the atoms at a given depth together. To approximate the work, we can divide the water in the tank into horizontal sections, approximate the volume of water in a section by a thin disk, and compute the amount of work required to lift each disk to the top of the tank. As usual, we take the limit as the sections get thinner and thinner to get the total work.

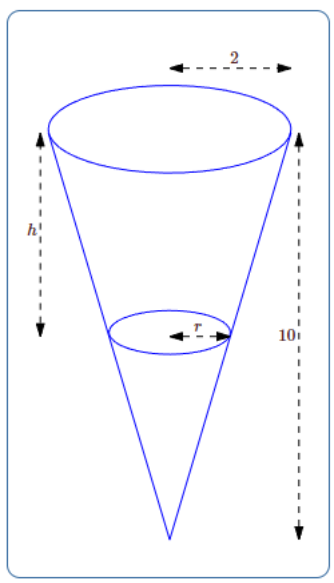

FIGURE 8. A conical water tank

At depth h the circular cross-section through the tank has radius $r = (10 - h)/5$, by similar triangles, and area $\pi(10 - h)^2/25$. A section of the tank at depth h thus has volume approximately $\pi(10 - h)^2/25\Delta h$ and so contains $\sigma\pi(10 - h)^2/25\Delta h$ kilograms of water, where σ is the density of water in kilograms per cubic meter; $\sigma \approx 1000$. The force due to gravity on this much water is $9.8\sigma\pi(10 - h)^2/25\Delta h$, and final1y, this section of

water molst be lifted a distance h, which requires $h9.8\sigma\pi(10-h)^2/25\Delta h$ Newton-meters of work. The total work is therefore

$$W = \frac{9.8\sigma\pi}{25}\int_0^{10} h(10-h)^2 dh = \frac{980000}{3}\pi \approx 1026254 \text{ Newton-meters.}$$

$\square$

3. Week 4 (3.15)

Problem 3.1 (8.1)**.** *Calculate the following improper integrals in infinite intervals.*

(1) $\displaystyle\int_1^{+\infty} \frac{\ln x}{(1+x)^2}\,\mathrm{d}x.$

(3) $\displaystyle\int_1^{+\infty} \frac{\mathrm{d}x}{x\sqrt{x^2+x+1}}.$

(5) $\displaystyle\int_0^{+\infty} \frac{\mathrm{d}x}{(1+x^2)^n}.$

Solution. (1)

$$\begin{aligned}
\int_1^{+\infty} \frac{\ln x}{(1+x)^2}\,\mathrm{d}x &= -\int_1^{+\infty} \ln x\ \mathrm{d}\left(\frac{1}{1+x}\right) \\
&= -\left.\frac{\ln x}{x+1}\right|_1^{+\infty} + \int_1^{+\infty} \frac{\mathrm{d}x}{x(x+1)} \\
&= -(0-0) + \ln\left(\frac{x}{x+1}\right)\Bigg|_1^{+\infty} \\
&= \ln 2.
\end{aligned}$$

(3)

$$\begin{aligned}
&\int_1^0 \frac{\mathrm{d}\left(\frac{1}{t}\right)}{\frac{1}{t}\sqrt{\frac{1}{t^2}+\frac{1}{t}+1}} = \int_0^1 \frac{\mathrm{d}t}{\sqrt{t^2+t+1}} \\
&= \int_0^1 \frac{\mathrm{d}t}{\sqrt{\left(t+\frac{1}{2}\right)^2+\frac{3}{4}}} \\
&= \ln\left[\left(t+\frac{1}{2}\right)+\sqrt{t^2+t+1}\right]\Bigg|_0^1 \\
&= \ln\frac{\sqrt{3}+\frac{3}{2}}{1+\frac{1}{2}} = \ln\left(\frac{2+\sqrt{3}}{\sqrt{3}}\right).
\end{aligned}$$

(5) **First way:**

$$
\begin{aligned}
I_n &= \int_0^{+\infty} \frac{\mathrm{d}x}{(1+x^2)^n} \\
&= \frac{x}{(1+x^2)^n}\bigg|_0^{+\infty} + \int_0^{+\infty} \frac{2nx^2}{(1+x^2)^{n+1}}\ \mathrm{d}x \\
&= 2n\left(I_n - I_{n+1}\right), \\
&\Rightarrow 2nI_{n+1} = (2n-1)I_n, \text{ i.e. } I_{n+1} = \frac{2n-1}{2n} I_n. \\
I_1 &= \int_0^{+\infty} \frac{\mathrm{d}x}{1+x^2} = \arctan x|_0^{+\infty} = \frac{\pi}{2} \\
I_n &= \frac{2n-3}{2n-2} \cdot \cdots \cdot \frac{1}{2} I_1 = \frac{\pi}{2} \cdot \frac{(2n-3)!!}{(2n-2)!!} \\
&\Rightarrow I_n = \begin{cases} \frac{\pi}{2}, n = 1 \\ \frac{(2n-3)!!}{(2n-2)!!}\frac{\pi}{2}, n \geq 2. \end{cases}
\end{aligned}
$$

Second way:

$$
\begin{aligned}
I_n &= \int_0^{+\infty} \frac{\mathrm{d}x}{(1+x^2)^n} = \int_0^{\frac{\pi}{2}} \frac{\mathrm{d}x}{\sec^{2n-2} x} \\
&= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x - \tan^2 x}{\sec^{2n-2} x}\ \mathrm{d}x \\
&= I_{n-1} - \int_0^{\frac{\pi}{2}} \frac{\tan^2 x}{\sec^{2n-2} x}\ \mathrm{d}x \\
&= I_{n-1} + \frac{1}{2n-2} \cdot \frac{\tan x}{\sec^{2n-2} x}\bigg|_0^{\frac{\pi}{2}} - \frac{1}{2n-2} I_{n-1} \\
&= \frac{2n-3}{2n-2} I_{n-1}, \quad n \geq 2.
\end{aligned}
$$

Then

$$
I_1 = \int_0^{+\infty} \frac{\mathrm{d}x}{1+x^2} = \frac{\pi}{2},
$$

and

$$
I_n = \frac{2n-3}{2n-2} \cdot \cdots \cdot \frac{1}{2} I_1 = \frac{\pi}{2} \cdot \frac{(2n-3)!!}{(2n-2)!!}, \quad n \geq 2.
$$

□

Problem 3.2 (8.2)**.** *Determine the convergence or divergence of the following improper intergals.*

(1) $\displaystyle\int_0^{+\infty} \frac{\sin x}{1+e^{-x}}\ \mathrm{d}x.$

(6) $\displaystyle\int_1^{+\infty} x\left(1-\cos\frac{1}{x}\right)^{\alpha}\ \mathrm{d}x,\ (\alpha>0).$

(7) $\displaystyle\int_1^{+\infty} \ln\left(\cos\frac{1}{x}+\sin^p\frac{1}{x}\right)\ \mathrm{d}x,\ (p>1).$

Hints. (1)

$$\int_{2N\pi}^{(2N+1)\pi} \frac{\sin x}{2}\ \mathrm{d}x \le \int_{2N\pi}^{(2N+1)\pi} \frac{\sin x}{1+e^{-x}}\ \mathrm{d}x$$

$$\Rightarrow \quad 1 \le \int_0^{(2N+1)\pi} \frac{\sin x}{1+e^{-x}}\ \mathrm{d}x - \int_0^{2N\pi} \frac{\sin x}{1+e^{-x}}\ \mathrm{d}x.$$

(6) Note that

$$1-\cos\frac{1}{x} = \frac{1}{2}\cdot\frac{1}{x^2}+o\left(\frac{1}{x^2}\right),\ x\to+\infty.$$

Then

$$x\left(1-\cos\frac{1}{x}\right)^{\alpha} \sim \frac{1}{2^{\alpha}}\cdot\frac{1}{x^{2\alpha-1}},\ x\to+\infty.$$

(7) Note that

$$\begin{aligned}\ln\left(\cos\frac{1}{x}+\sin^p\frac{1}{x}\right) &= \ln\left(1+\cos\frac{1}{x}-1+\sin^p\frac{1}{x}\right)\\ &= \cos\frac{1}{x}-1+\sin^p\frac{1}{x}+o\left(\frac{1}{x^2}\right)+o\left(\frac{1}{x^p}\right)\\ &= -\frac{1}{2}\cdot\frac{1}{x^2}+\frac{1}{x^p}+o\left(\frac{1}{x^2}\right)+o\left(\frac{1}{x^p}\right),\ x\to+\infty.\end{aligned}$$

□

Problem 3.3 (8.3)**.** *Determine the convergence or absolute convergence of the following improper intergals.*

(3) $\displaystyle\int_1^{+\infty} \frac{\sin x}{x^{\alpha}+\sin x}\ \mathrm{d}x.$

(4) $\displaystyle\int_1^{+\infty} \sin\left(\frac{\sin x}{x}\right)\ \mathrm{d}x.$

Hints. (3)

$$\int_1^{+\infty} \frac{\sin x}{x^{\alpha}+\sin x}\ \mathrm{d}x = \int_1^{+\infty} \frac{\sin x}{x^{\alpha}}\ \mathrm{d}x - \int_1^{+\infty} \frac{\sin^2 x}{x^{\alpha}(x^{\alpha}+\sin x)}\ \mathrm{d}x.$$

(4)

$$\sin\left(\frac{\sin x}{x}\right) = \frac{\sin x}{x}+o\left(\frac{1}{x^3}\right),\ x\to+\infty.$$

$\square$

Problem 3.4 (8.9)**.** *Suppose that $f(x) \in C(-\infty, +\infty)$ is a periodic function with period 2π, and $\int_0^{2\pi} f(x)\ \mathrm{d}x = 0$. Prove that for any $\alpha > 0$, the improper intergral $\int_1^{+\infty} x^{-\alpha} f(x)\ \mathrm{d}x$ converges.*

Proof. Since $f(x)$ is a continuous periodic function on $\mathbb{R}$, we know that its intergral exists on any finite closed intervals. Note that $\forall\, X > 0$, there exist $k \in \mathbb{Z}$ and $0 \le r < 2\pi$ such that

$$X = 2k\pi + r.$$

Then

$$\begin{aligned}
\left|\int_1^X f(x)\ \mathrm{d}x\right| &\le \left|\int_0^1 f(x)\ \mathrm{d}x\right| + \left|\int_0^X f(x)\ \mathrm{d}x\right| \\
&= \left|\int_0^1 f(x)\ \mathrm{d}x\right| + \left|\int_0^{2k\pi+r} f(x)\ \mathrm{d}x\right| \\
&= \left|\int_0^1 f(x)\ \mathrm{d}x\right| + \left|\int_0^r f(x)\ \mathrm{d}x\right| \\
&\le 2\int_0^{2\pi} |f(x)|\ \mathrm{d}x.
\end{aligned}$$

Hence by the Dirichlet test rule, we know that the improper intergral $\int_1^{+\infty} x^{-\alpha} f(x)\ \mathrm{d}x$ converges. $\square$

Problem 3.5 (8.10)**.** *Suppose that $f(x)$ is uniformly continuous on $[0, +\infty$, and the improper intergral $\int_0^{+\infty} f(x)\ \mathrm{d}x$ converges. Prove that $\lim_{x\to+\infty} f(x) = 0$.*

Proof. Prove by contradiction. Assume that $\lim_{x\to+\infty} f(x) \ne 0$, i.e. there exists $\varepsilon_0 > 0$, such that $\forall\, n \in \mathbb{N}$, there is $x_n > n$ satisfying $|f(x_n)| \ge \varepsilon_0$. Since $f(x)$ is uniformly continuous, we have that for $\varepsilon_0/2$, there exists a $\delta_0 > 0$, such that $\forall\, y : |y - x_n| < \delta_0$,

$$|f(y) - f(x_n)| < \frac{\varepsilon_0}{2}.$$

Then

$$|f(y)| \ge |f(x_n)| - |f(y) - f(x_n)| > \frac{\varepsilon_0}{2},\ \forall\, y \in (x_n - \delta_0, x_n + \delta_0),$$

and $f(y)$, $f(x_n)$ have the same sign in $(x_n - \delta_0, x_n + \delta_0)$. Therefore, we have

$$\left|\int_{x_n - \frac{\delta_0}{2}}^{x_n + \frac{\delta_0}{2}} f(x)\ \mathrm{d}x\right| \ge \frac{\varepsilon_0 \delta_0}{2},$$

contradicts with the convergence of $\int_0^{+\infty} f(x)\ \mathrm{d}x$. Thus, $\lim_{x\to+\infty} f(x) = 0$. $\square$

Problem 3.6 (8.14)**.** *Calculate the following improper integrals with discontinuous integrand.*

(1) $\int_0^1 \frac{\mathrm{d}x}{(2-x)\sqrt{1-x}}$.

(6) $\int_0^{\frac{\pi}{2}} \sqrt{\tan x}\ \mathrm{d}x$.

Solution. (1)

$$\int_0^1 \frac{\mathrm{d}x}{(2-x)\sqrt{1-x}} = -2\int_0^1 \frac{\mathrm{d}\sqrt{1-x}}{2-x} = \frac{\pi}{2}.$$

(6)

$$I = \int_0^{\pi/2} \sqrt{\tan x}\ \mathrm{d}x = \int_0^{\pi/2} \frac{\sin x}{\sqrt{\cos x \sin x}}\ \mathrm{d}x = \int_0^{\pi/2} \frac{\cos x}{\sqrt{\cos x \sin x}}\ \mathrm{d}x.$$

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}\ \mathrm{d}x \\ &= \int_0^{\pi/2} \frac{\sqrt{2}\ \mathrm{d}(\sin x - \cos x)}{\sqrt{1-(\sin x - \cos x)^2}} \\ &= \sqrt{2}\pi. \end{aligned}$$

i.e. $I = \frac{\pi}{\sqrt{2}}$. $\square$

Problem 3.7 (8.16)**.** *Determine the convergence or divergence of the following improper intergals.*

(5) $\int_0^{+\infty} \frac{\sin\left(x + \frac{1}{x}\right)}{x^p}\ \mathrm{d}x$.

Hints. Note that

$$\frac{\sin\left(x + \frac{1}{x}\right)}{x^p} = \frac{\sin x \cos\frac{1}{x}}{x^p} + \frac{\cos x \sin\frac{1}{x}}{x^p}.$$

$\square$

Problem 3.8. *Suppose that $f(x)$ is integrable on any finite closed interval, and $\lim_{x\to+\infty} f(x) = A$, $\lim_{x\to-\infty} f(x) = B$. Prove that $\forall\, a > 0$, improper inegral*

$$\int_{-\infty}^{+\infty} [f(x+a) - f(x)]\ \mathrm{d}x$$

converges.

Hints. For any M, N, we have

$$\begin{aligned}\int_M^N [f(x+a)-f(x)]\ \mathrm{d}x &= \int_M^N f(x+a)\ \mathrm{d}x - \int_M^N f(x)\ \mathrm{d}x \\ &= \int_{M+a}^{N+a} f(x)\ \mathrm{d}x - \int_M^N f(x)\ \mathrm{d}x \\ &= \int_N^{N+a} f(x)\ \mathrm{d}x - \int_M^{M+a} f(x)\ \mathrm{d}x \\ &\to a(A-B), \quad \text{as } M\to-\infty, N\to+\infty.\end{aligned}$$

□

Problem 3.9. *Suppose that $f(x)$ is integrable on any finite closed interval, and $\int_{-\infty}^{+\infty} f^2\ \mathrm{d}x$ converges. Prove that $\forall\, a>0$, improper inegral*

$$\int_{-\infty}^{+\infty} |f(x+a)f(x)|\ \mathrm{d}x$$

converges.

Hints. For any M, N, by Cauchy-Schwarz's inequality, we have

$$\begin{aligned}\left[\int_M^N |f(x+a)f(x)|\ \mathrm{d}x\right]^2 &\le \int_M^N [f(x+a)]^2\ \mathrm{d}x \cdot \int_M^N [f(x)]^2\ \mathrm{d}x \\ &= \int_{M+a}^{N+a} [f(x)]^2\ \mathrm{d}x \cdot \int_M^N [f(x)]^2\ \mathrm{d}x.\end{aligned}$$

Then letting $M\to-\infty, N\to+\infty$. □

Problem 3.10. *Prove that $\int_0^{+\infty} \frac{x}{1+x^6\sin^2 x}\ \mathrm{d}x$ converges.*

Hints. It suffices to show that

$$F(A) = \int_0^A \frac{x}{1+x^6\sin^2 x}\ \mathrm{d}x$$

is bounded on $[0,+\infty)$. Note that

$$\begin{aligned}\int_0^{n\pi} \frac{x}{1+x^6\sin^2 x}\ \mathrm{d}x &= \sum_{k=1}^n \int_{(k-1)\pi}^{k\pi} \frac{x}{1+x^6\sin^2 x}\ \mathrm{d}x \\ &\le \sum_{k=1}^n k\pi \int_{(k-1)\pi}^{k\pi} \frac{\mathrm{d}x}{1+(k-1)^6\pi^6\sin^2 x} \\ &= \sum_{k=1}^n k\pi \int_0^{\pi} \frac{\mathrm{d}x}{1+(k-1)^6\pi^6\sin^2 x}\end{aligned}$$

$$
\begin{aligned}
&= \sum_{k=1}^{n} 2k\pi \int_0^{\pi/2} \frac{\mathrm{d}x}{1+(k-1)^6\pi^6\sin^2 x} \\
&\le \sum_{k=1}^{n} 2k\pi \int_0^{\pi/2} \frac{\mathrm{d}x}{1+4(k-1)^6\pi^4 x^2} \\
&= \sum_{k=1}^{n} \frac{k}{\pi(k-1)^3} \int_0^{(k-1)^3\pi^3} \frac{\mathrm{d}x}{1+x^2} \\
&\sim \sum_{k=1}^{n} \frac{1}{2k^2} \quad (k \to \infty).
\end{aligned}
$$

Hence $F(n\pi)$ is bounded, which gives us that $F(A)$ converges. □

Problem 3.11. *Suppose that $f(x)$ is continuous on $[0,+\infty)$, and $\lim_{x\to+\infty} f(x)$ exists. For $0<a<b$, calculate*

$$
\int_0^{+\infty} \frac{f(ax)-f(bx)}{x}\,\mathrm{d}x.
$$

Hints. For $0<r<R<+\infty$, we have

$$
\begin{aligned}
\int_r^R \frac{f(ax)-f(bx)}{x}\,\mathrm{d}x &= \int_r^R \frac{f(ax)}{x}\,\mathrm{d}x - \int_r^R \frac{f(bx)}{x}\,\mathrm{d}x \\
&= \int_{ar}^{aR} \frac{f(x)}{x}\,\mathrm{d}x - \int_{br}^{bR} \frac{f(x)}{x}\,\mathrm{d}x \\
&= \int_{ar}^{br} \frac{f(x)}{x}\,\mathrm{d}x - \int_{aR}^{bR} \frac{f(x)}{x}\,\mathrm{d}x.
\end{aligned}
$$

Then by the first mean value theorem for definite integrals, we have

$$
\begin{aligned}
\int_{ar}^{br} \frac{f(x)}{x}\,\mathrm{d}x &= f(\xi)\int_{ar}^{br} \frac{\mathrm{d}x}{x} = f(\xi)\ln\frac{b}{a} \quad (ar<\xi<br), \\
\int_{aR}^{bR} \frac{f(x)}{x}\,\mathrm{d}x &= f(\eta)\int_{ar}^{br} \frac{\mathrm{d}x}{x} = f(\eta)\ln\frac{b}{a} \quad (aR<\eta<bR).
\end{aligned}
$$

Then letting $r\to 0$, and $N\to+\infty$ yield

$$
\int_0^{+\infty} \frac{f(ax)-f(bx)}{x}\,\mathrm{d}x = [f(0)-f(+\infty)]\ln\frac{b}{a}.
$$

□

Problem 3.12. *Prove that for any $\alpha\in\mathbb{R}$, there ia*

$$
\int_0^{+\infty} \frac{\mathrm{d}x}{(1+x^2)(1+x^\alpha)} = \frac{\pi}{4}
$$

converges.

Hints. Note that by changing of variable, we have

$$\int_0^1 \frac{\mathrm{d}x}{(1+x^2)(1+x^\alpha)} = \int_1^{+\infty} \frac{x^\alpha \, \mathrm{d}x}{(1+x^2)(1+x^\alpha)}.$$

□

Problem 3.13. *Suppose that $f(x)$ is integrable on any finite closed interval, and $\forall\, p \geq 1$, $\int_{-\infty}^{+\infty} |f|^p \, \mathrm{d}x$ converges. Prove that*

$$\lim_{h\to 0} \int_{-\infty}^{+\infty} |f(x+h) - f(x)|^p \, \mathrm{d}x = 0.$$

Hints. Note that

$$\int_{-\infty}^{+\infty} |f(x+h) - f(x)|^p \, \mathrm{d}x = \int_{-\infty}^{-R} |f(x+h) - f(x)|^p \, \mathrm{d}x + \int_{R}^{+\infty} |f(x+h) - f(x)|^p \, \mathrm{d}x + \int_{-R}^{R} |f(x+h) - f(x)|^p \, \mathrm{d}x.$$

Since $\int_{-\infty}^{+\infty} |f|^p \, \mathrm{d}x$ converges, we know that for $\forall\, \varepsilon > 0$, there exists $R > 0$ such that

$$\int_{-\infty}^{-R} |f(x+h) - f(x)|^p \, \mathrm{d}x < \frac{\varepsilon}{3},$$

and

$$\int_{R}^{+\infty} |f(x+h) - f(x)|^p \, \mathrm{d}x < \frac{\varepsilon}{3}.$$

Since

$$\lim_{h\to 0} \int_{-R}^{R} |f(x+h) - f(x)|^p \, \mathrm{d}x = 0,$$

we have

$$\int_{-\infty}^{+\infty} |f(x+h) - f(x)|^p \, \mathrm{d}x < \varepsilon.$$

□

4. Week 5 (3.22)

Problem 4.1 (8.17)**.** *Determine the conditional convergence or absolute convergence of the following improper intergals.*

(1) $\displaystyle\int_0^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x.$

Hints. (1) When $p \geq 0$, the flaw points are 0 and $+\infty$. Then

$$\int_0^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x = \int_0^1 |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x + \int_1^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x.$$

Note that

$$\frac{\sin x}{x^q} \sim \frac{1}{x^{q-1}}, \text{ as } x \to 0.$$

Hence when $1 < q < 2$, $\displaystyle\int_0^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x$ converges absolutely; when $0 < q < 1$, $\displaystyle\int_0^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x$ converges conditionally.

When $p < 0$, we know that 1 is also the flaw point, and there is

$$|\ln x|^p \sim \frac{1}{|x-1|^{-p}}, \text{ as } x \to 1.$$

Hence for $-1 < p < 0$, we write

$$\int_0^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x = \int_0^{1/2} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x + \int_{1/2}^{3/2} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x + \int_{3/2}^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x.$$

Hence when $1 < q < 2$, $\displaystyle\int_0^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x$ converges absolutely; when $0 \leq q < 1$, $\displaystyle\int_0^{+\infty} |\ln x|^p \frac{\sin x}{x^q}\ \mathrm{d}x$ converges conditionally. □

Problem 4.2 (8.21)**.** *Suppose that $f(x)$ is monotonic on $(0, 1]$, and $\displaystyle\int_0^1 f(x)\ \mathrm{d}x$ converges. Prove that*

$$\int_0^1 f(x)\ \mathrm{d}x = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f\left(\frac{k}{n}\right).$$

If we remove the condition that $f(x)$ is monotonic on $(0, 1]$, is the conclusion still true?

Proof. Without loss of generality, we may assume that $f(x)$ is monotonic decreasing on $(0,1]$. Hence there is

$$\int_{\frac{1}{n}}^{1} f(x)\ \mathrm{d}x + \frac{f(1)}{n} \leq \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) \leq \int_0^1 f(x)\ \mathrm{d}x,$$

which gives us that

$$\int_0^1 f(x)\ \mathrm{d}x = \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right).$$

If $f(x)$ is not monotonic decreasing on $(0,1]$, then the conclusion may be wrong. For example, we can consider the function

$$f(x) = \begin{cases} n^2, & x = \dfrac{1}{n}, \\ 0, & others. \end{cases}$$

□

Problem 4.3 (8.22)**.** *Suppose that $f(x)$ is monotonic on $[0,+\infty)$, and $\int_0^{+\infty} f(x)\ \mathrm{d}x$ converges. Prove that*

$$\lim_{\lambda\to\infty} \int_0^{+\infty} f(x)\sin\lambda x\ \mathrm{d}x = 0.$$

Hint. Consider

$$\int_0^{+\infty} f(x)\sin\lambda x\ \mathrm{d}x = \int_0^R f(x)\sin\lambda x\ \mathrm{d}x + \int_R^{+\infty} f(x)\sin\lambda x\ \mathrm{d}x,$$

and use Exercise 12.14 in the notes of *Mathematical analysis I*. □

Problem 4.4 (8.24)**.** *Suppose that $f(x)$ is monotonic on $[0,+\infty)$, and $g(x) \not\equiv 0$ is periodic and continuous on $\mathbb{R}$ with period $T > 0$. Prove that $\int_0^{+\infty} f(x)\ \mathrm{d}x$ converges if and only if $\int_0^{+\infty} f(x)|g(x)|\ \mathrm{d}x$ converges.*

Proof. "$\Longrightarrow$" Since $f(x)$ is monotonic on $[0,+\infty)$, we may assume that $f(x) \geq 0$. Since $g(x$ is periodic and continuous on $\mathbb{R}$, we know that there exists a $M > 0$ such that $|g(x)| \leq M$, $\forall\, x \in \mathbb{R}$. Then, we have

$$\int_0^{+\infty} f(x)|g(x)|\ \mathrm{d}x \leq M\int_0^{+\infty} f(x)\ \mathrm{d}x < \infty.$$

"$\Longleftarrow$" Since $g(x) \not\equiv 0$ is a continuous and periodic function with period $T > 0$, we know there exist a $A > 0$ and $[a,b] \subset (0,T)$ such that for any $x \in [a,b]$, there is $|g(x)| > A$. Then we know

$$\int_0^T |g(x)|\,\mathrm{d}x \geq \int_a^b |g(x)|\,\mathrm{d}x \geq A(b-a).$$

Without loss of generality, we may assume that $f(x)$ is monotonic decreasing and non-negative. Then

$$\int_{(k-1)T}^{kT} f(x)|g(x)|\ \mathrm{d}x \geq f(kT)\int_{(k-1)T}^{kT} |g(x)|\ \mathrm{d}x \geq A(b-a)f(kT), \quad k \geq 1.$$

Note that

$$\int_{kT}^{(k+1)T} f(x)\ \mathrm{d}x \leq f(kT)T \leq \frac{T}{A(b-a)}\int_{(k-1)T}^{kT} f(x)|g(x)|\ \mathrm{d}x.$$

Then

$$\begin{aligned}\int_{T}^{(n+1)T} f(x)\ \mathrm{d}x &= \sum_{k=1}^{n}\int_{kT}^{(k+1)T} f(x)\ \mathrm{d}x \\ &\leq \frac{T}{A(b-a)}\sum_{k=1}^{n}\int_{(k-1)T}^{kT} f(x)|g(x)|\ \mathrm{d}x \\ &= \frac{T}{A(b-a)}\int_{0}^{nT} f(x)|g(x)|\ \mathrm{d}x \\ &\to \int_{0}^{+\infty} f(x)|g(x)|\ \mathrm{d}x, \quad \text{as } n \to \infty,\end{aligned}$$

which implies that $\displaystyle\int_{0}^{+\infty} f(x)\ \mathrm{d}x$ converges. □

Problem 4.5. *Suppose that $f(x)$ is differentiable, $\displaystyle\int_{0}^{+\infty} f(x)\ \mathrm{d}x$ and $\displaystyle\int_{0}^{+\infty} f'(x)\ \mathrm{d}x$ converge. Prove that $\displaystyle\lim_{x\to+\infty} f(x) = 0$.*

Hint. Firstly, using $\displaystyle\int_{0}^{+\infty} f'(x)\ \mathrm{d}x$ converges to prove $\displaystyle\lim_{x\to+\infty} f(x) = \alpha$ exists (by Cauchy's convergence test and Heine's theorem). Then, using $\displaystyle\int_{0}^{+\infty} f(x)\ \mathrm{d}x$ to show $\alpha = 0$ (by contradiction). □

Exercise 4.6. *Suppose that $F(x) = \displaystyle\int_{0}^{x}\left(\frac{1}{t} - \left[\frac{1}{t}\right]\right)\ \mathrm{d}t$. Prove that $F'(0) = \dfrac{1}{2}$.*

Hints: By changing of variable, $t = \frac{x}{y}$, we have

$$F(x) = \int_{0}^{x}\left(\frac{1}{t} - \left[\frac{1}{t}\right]\right)\ \mathrm{d}t = x\int_{1}^{+\infty}\left(\frac{y}{x} - \left[\frac{y}{x}\right]\right)\frac{1}{y^2}\ \mathrm{d}y.$$

Hence

$$\frac{F(x)}{x} = \int_{1}^{+\infty}\left(\frac{y}{x} - \left[\frac{y}{x}\right]\right)\frac{1}{y^2}\ \mathrm{d}y = \int_{1}^{+\infty}\frac{\varphi\left(\frac{y}{x}\right)}{y^2}\ \mathrm{d}y,$$

where $\varphi(t) = t - [t]$ is a periodic function with period $T = 1$. Hence, by Riemann's theorem, we know

$$\begin{aligned} F'(0) &= \lim_{x \to 0} \frac{F(x)}{x} = \frac{1}{T} \int_0^T \varphi(y) \, \mathrm{d}y \cdot \int_1^{+\infty} \frac{1}{y^2} \, \mathrm{d}y \\ &= \int_0^1 y \, \mathrm{d}y = \frac{1}{2}. \end{aligned}$$

$\square$

5. Week 6 (3.29)

Problem 5.1 (9.10)**.** *Discuss the convergence and divergence of the following series.*

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{[\sqrt{n}]}}{n^p}$, $(p > 0)$.

Problem 5.2 (9.16)**.** *Suppose that* $\displaystyle\sum_{n=1}^{\infty} a_n$ *converges, and* $\displaystyle\sum_{n=1}^{\infty}(b_n - b_{n+1})$ *converges absolutely. Prove that* $\displaystyle\sum_{n=1}^{\infty} a_n b_n$ *converges.*

Problem 5.3 (9.21)**.** *Suppose that the sequence* $\{a_n\}$ *satisfies* $\lim\limits_{n\to\infty} na_n$ *exists and series* $\displaystyle\sum_{n=1}^{\infty} n(a_n - a_{n+1})$ *converges. Prove that series* $\displaystyle\sum_{n=1}^{\infty} a_n$ *converges.*

Problem 5.4 (9.23)**.** *Suppose that non-constant function* $f(x)$ *is nonnegative and continuous on* $[0,1]$*, and* $f(x) \le 1$*,* $x \in [0,1]$*. For* $\forall\, n \in \mathbb{N}$*, defining* $a_n = \left[\displaystyle\int_0^1 f(x)\,\mathrm{d}x\right]^{\frac{1}{n}}$*. Prove that series* $\displaystyle\sum_{n=1}^{\infty}(1 - a_n)$ *converges.*

Problem 5.5. *Suppose that* $a_n > 0$ *and* $\displaystyle\sum_{n=1}^{\infty} \frac{1}{a_n}$ *converges. Prove that*

$$\sum_{n=1}^{\infty} \frac{n^2}{(a_1 + a_2 + \cdots + a_n)^2} a_n$$

converges, too.

Problem 5.6. *Suppose that* $x_n = 1 + \dfrac{1}{\sqrt{2}} + \cdots + \dfrac{1}{\sqrt{n}} - 2\sqrt{n}$*. Prove that* $\lim\limits_{n\to\infty} x_n$ *exists.*

Problem 5.7. *Suppose that* $a_n > 0$ *and* $S = \displaystyle\sum_{n=1}^{\infty} a_n$ *converges. Prove that*

$$\lim_{n\to\infty} \frac{\sum\limits_{k=1}^{n} k a_k}{n} = 0,$$

and

$$\sum_{n=1}^{\infty} \frac{a_1 + 2a_2 + \cdots + na_n}{n(n+1)} = S.$$

Problem 5.8. *Suppose that* $a_n > 0$, $\sum_{n=1}^{\infty} a_n$ *converges and* $\{a_n - a_{n+1}\}$ *is decreasing. Prove that* $\{a_n\}$ *is decreasing and*

$$\lim_{n\to\infty} \frac{1}{a_{n+1}} - \frac{1}{a_n} = +\infty.$$

Problem 5.9. *Suppose that* $\{a_n\}$ *is positive and decreasing. If series* $\sum_{n=1}^{\infty} \frac{a_n}{\sqrt{n}}$ *converges, prove that* $\sum_{n=1}^{\infty} a_n^2$ *also converges.*

Problem 5.10.

(1) *If for evrey sequence* $\{b_n\}$ *satisfying* $\lim_{n\to\infty} b_n = 0$, *there is* $\sum_{n=1}^{\infty} a_n b_n$ *converges. Prove that* $\sum_{n=1}^{\infty} a_n$ *converges absolutely.*

(2) *If for every convergent series* $\sum_{n=1}^{\infty} b_n$, *we have that series* $\sum_{n=1}^{\infty} a_n b_n$ *converges. Prove that* $\sum_{n=1}^{\infty} |a_n - a_{n+1}|$ *must be convergent.*

6. Week 7 (4.5)

Problem 6.1. *Suppose that $\{x_n\}$ is positive, monotonic decreasing and $\sum_{n=1}^{\infty} x_n = +\infty$. Prove that*

$$\sum_{n=1}^{\infty} x_n e^{-\frac{x_n}{x_{n+1}}} = +\infty.$$

Proof. **Firsr way:** Clearly, $\{x_n\}$ converges. If $x_n \to a > 0$, then $x_n/x_{n+1} \to 1$, and hence $\mathrm{e}^{-x_n/x_{n+1}}$ lower bounded, say by $b > 0$, and hence

$$\sum x_n \mathrm{e}^{-x_n/x_{n+1}} \geq b \sum x_n = \infty.$$

If $x_n \to 0$, set

$$S_k = \{n \in \mathbb{N} : 2^{-k} < x_n \leq 2^{-k+1}\}$$

Then $\sum_{k=0}^{\infty} \sum_{n \in S_k} x_n = \infty$, and

$$\sum_{k=0}^{\infty} 2^{-k+1}|S_k| \geq \sum_{k=0}^{\infty} \sum_{n \in S_k} x_n \geq \sum_{k=0}^{\infty} 2^{-k}|S_k|.$$

Hence $\sum_{k=0}^{\infty} 2^{-k}|S_k| = \frac{1}{2}\sum_{k=0}^{\infty} 2^{-k+1}|S_k| = \infty$. Also, if $n, n+1 \in S_k$, then $\mathrm{e}^{-x_n/x_{n+1}} \geq \mathrm{e}^{-2}$. Hence, if $S_k = \{j, j+1, \ldots, \ell\}$, then

$$\sum_{n \in S_k} x_n \mathrm{e}^{-x_n/x_{n+1}} > \sum_{n=j}^{\ell-1} x_n \mathrm{e}^{-x_n/x_{n+1}} \geq \sum_{n=j}^{\ell-1} x_n \mathrm{e}^{-2} \geq (|S_k| - 1)\mathrm{e}^{-2}.$$

Thus

$$\sum_{k=0}^{\infty} \sum_{n \in S_k} x_n \mathrm{e}^{-x_n/x_{n+1}} \geq \sum_{k=0}^{\infty} 2^{-k+1}(|S_k| - 1)\mathrm{e}^{-2} = \infty$$

Second way: Divide $\mathbb{N}$ into two complementary sets of indices

$$A = \{n \in \mathbb{N} \mid x_{n+1} \leq \frac{1}{2}x_n\}, B = \{n \in \mathbb{N} \mid x_{n+1} > \frac{1}{2}x_n\}.$$

If $k < l$ are two elements of A then $x_l \leq \frac{1}{2}x_k$ (here we need the fact the the given sequence is decreasing). It follows that the k'th element of A is $\leq 2^{-k}x_1$. A can be empty, finite, or infinite, but in any case is

$$\sum_{n \in A} x_n < \infty$$

and therefore

$$\sum_{n \in B} x_n = \infty.$$

Then

$$\sum_{n=1}^{\infty} x_n e^{-\frac{x_n}{x_{n+1}}} \geq \sum_{n \in B} x_n e^{-\frac{x_n}{x_{n+1}}} \geq e^{-2} \sum_{n \in B} x_n = \infty.$$

$\square$

Remark 6.2. *The proof shows that under the given conditions,*

$$\sum_{n=1}^{\infty} x_n f\left(\frac{x_n}{x_{n+1}}\right) = \infty$$

holds for any positive, monotonic decreasing function $f : (0, \infty) \to (0, \infty)$.

Problem 6.3. *Let* f *be an increasing function on* $[0, 1]$ *such that* $0 \leq f(x) \leq 1$ *and* $\int_0^1 (f(x) - x)dx = 0$. *Show that*

$$\int_0^1 |f(x) - x|dx \leq \frac{1}{2}.$$

Proof. Here we present a bit different, calculus-themed approach. In this answer, we will assume that $f : [0, 1] \to [0, 1]$ is monotone-increasing. We also write $I(f) = \int_0^1 |f(x) - x| \, \mathrm{d}x$ for brevity.

Step 1 - Proof under extra assumptions. Assume further that f is piecewise-smooth, $f(0) = 0$, and $f(1) = 1$. Then by the formula $\int |x| \, \mathrm{d}x = \frac{1}{2}x|x| + \mathsf{C}$, we have

$$\int_0^1 |f(x) - x|(f'(x) - 1) \, \mathrm{d}x = \left[\frac{1}{2}|f(x) - x|(f(x) - x)\right]_0^1 = 0.$$

In particular,

$$I(f) = \frac{1}{2}\int_0^1 |f(x) - x|(f'(x) + 1) \, \mathrm{d}x.$$

Now pick $\alpha \in [0, 1]$ so that $f(\alpha) + \alpha = 1$. (This is possible since $x \mapsto f(x) + x$ increases from 0 to 2. Then by triangle inequality,

$$\int_0^\alpha |f(x) - x|(f'(x) + 1) \, \mathrm{d}x \leq \int_0^\alpha (f(x) + x)(f'(x) + 1) \, \mathrm{d}x = \frac{1}{2}.$$

Similarly, by writing $|f(x) - x| = |(1 - f(x)) - (1 - x)| \leq (1 - f(x)) + (1 - x)$, we get

$$\int_\alpha^1 |f(x) - x|(f'(x) + 1) \, \mathrm{d}x \leq \int_\alpha^1 (2 - f(x) - x)(f'(x) + 1) \, \mathrm{d}x = \frac{1}{2}.$$

Therefore $\int_0^1 |f(x) - x|(f'(x) + 1) \, \mathrm{d}x \leq 1$, which in turn implies $I(f) \leq \frac{1}{2}$ as required.

Step 2 - General case. For the general case, let f_n be the linear interpolation of the points

$$(0, 0), \quad (\tfrac{1}{n}, f(\tfrac{1}{n})), \quad \cdots, \quad (\tfrac{n-1}{n}, f(\tfrac{n-1}{n})).$$

Then by monotonicity,

$$\begin{aligned} |I(f_n) - I(f)| &\leq \int_0^1 |f_n(x) - f(x)| \, \mathrm{d}x = \sum_{k=1}^{n} \int_{\frac{k-1}{n}}^{\frac{k}{n}} |f_n(x) - f(x)| \, \mathrm{d}x \\ &\leq \frac{1}{n}\left([f(\tfrac{1}{n}) - 0] + \sum_{k=2}^{n-1}[f(\tfrac{k}{n}) - f(\tfrac{k-1}{n})] + [1 - f(\tfrac{n-1}{n})]\right) \end{aligned}$$

$$= \frac{1}{n},$$

hence $I(f_n) \to I(f)$ as $n \to \infty$ and the desired inequality $I(f) \leq \frac{1}{2}$ follows from the previous step. $\square$

Remark 6.4. *Let $\gamma(t) = (f(t)+t, f(t)-t)$. Then $\int_0^1 |f(t)-t|(f'(t)+1)\,\mathrm{d}t = \int_\gamma |y|\,\mathrm{d}x$ computes the area between the path γ and the horizontal axis. Note that γ is essentially the $-45°$-rotation of the graph $y = f(x)$ up to scaling.*

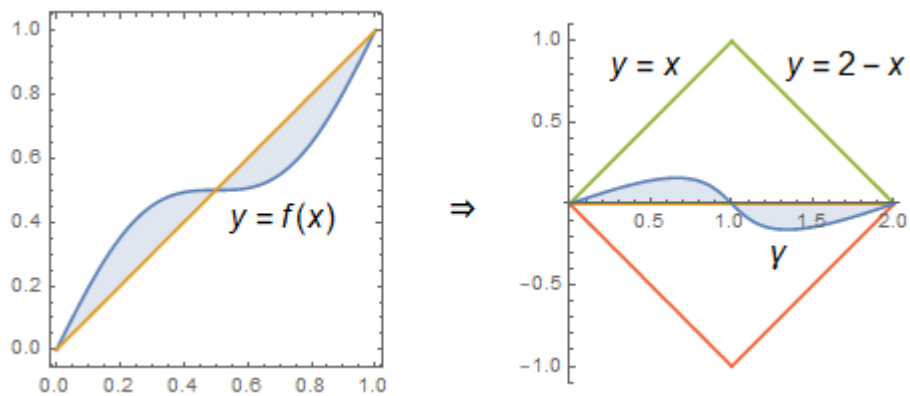

FIGURE 9. Graph

Then the above bounds immediately follow from the fact that the graph of γ defines a function on $[0,2]$ which is squeezed between lines $y = \pm x$ and $y = \pm(2-x)$.

Problem 6.5. *Let f be an increasing function on $[0,1]$ such that $0 \leq f(x) \leq 1$ and $\int_0^1 (f(x)-x)dx = 0$. Show that*

$$\int_0^1 |f(x)-x|dx \leq \frac{1}{4}.$$

Proof. **Step 1. (Proof under extra assumptions)** Assume that

- $f : [0,1] \to [0,1]$ is continuous and non-decreasing;
- $\displaystyle\int_0^1 (f(x)-x)\,\mathrm{d}x = 0$;
- $f(0) = 0$, and $f(1) = 1$.

Then the set $U_+ = \{x \in [0,1] : f(x) > x\}$ is open, hence it is written as the union of at most countably many disjoint open intervals (a_i, b_i), $i = 1, 2, \ldots$ Also, the continuity of f forces that $f(b_i) = b_i$. So,

$$I_+ := \int_{U_+} |f(x)-x|\,\mathrm{d}x = \sum_i \int_{(a_i,b_i)} (f(x)-x)\,\mathrm{d}x$$

$$\leq \sum_i \int_{(a_i,b_i)} (b_i - x)\,\mathrm{d}x = \sum_i \frac{(b_i - a_i)^2}{2}$$
$$\leq \frac{1}{2}\left(\sum_i (b_i - a_i)\right)^2 = \frac{1}{2}|U_+|^2.$$

A similar argument shows that, for $U_- = \{x \in [0,1] : f(x) < x\}$ we have

$$I_- := \int_{U_-} |f(x) - x|\,\mathrm{d}x \leq \frac{1}{2}|(U_-|^2.$$

Moreover, from $\int_0^1 (f(x) - x)\,\mathrm{d}x = 0$ we get $I_+ = I_-$. Therefore, together with the observations $\int_0^1 |f(x) - x|\,\mathrm{d}x = 2I_+ = 2I_-$ and $|U_+| + |U_-| \leq 1$, we conclude that

$$\int_0^1 |f(x) - x|\,\mathrm{d}x \leq \min\{|U_+|, |U_-|\}^2 \leq \frac{1}{4}.$$

These inequalities have a nice interpretation in terms of areas:

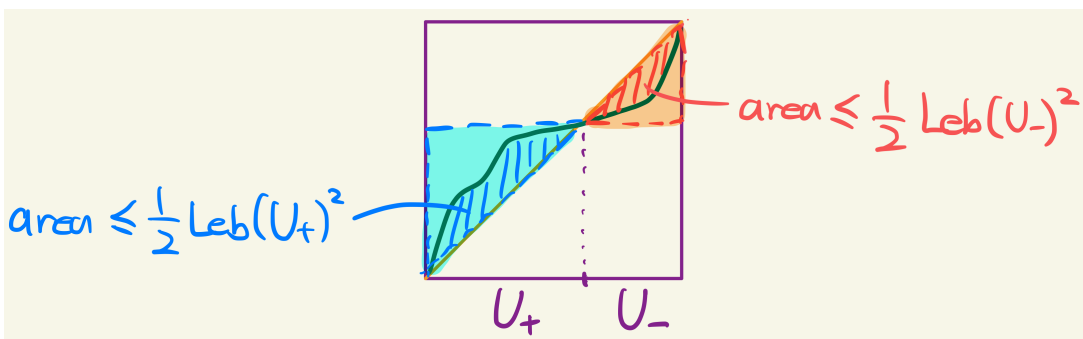

FIGURE 10. Graph

Step 2. (General case by approximation) Now suppose $f : [0,1] \to [0,1]$ is non-decreasing and satisfies $\int_0^1 (f(x) - x)\,\mathrm{d}x = 0$. Then it is not hard to find a sequence $f_n(x)$ satisfying the conditions in Step 1 and $f_n(x) \to f(x)$ for almost every x. So, by the dominated convergence theorem,

$$\int_0^1 |f(x) - x|\,\mathrm{d}x = \lim_{n\to\infty} \int_0^1 |f_n(x) - x|\,\mathrm{d}x \leq \frac{1}{4}.$$

□

Actually, there is an extension of the above two problems. But the proof needs some tools of *real analysis*. We still give proof here in case someone is interested in. Note that in this case, the bound of RHS can not be improved to 1/4.

Exercise 6.6 (Hard!). *f, g are monotonically increasing in $[0,1]$ and $0 \le f, g \le 1$. $\int_0^1 (f-g)\mathrm{d}x = 0$. Prove that*

$$\int_0^1 |f-g|\mathrm{d}x \le \frac{1}{2}.$$

Hint: Let $f = \mathbf{1}_{[\frac{1}{2},1]}$, $g = \frac{1}{2}$, then $\int_0^1 |f-g|\mathrm{d}x = \frac{1}{2}$. Except for swapping f, g, this is the only case to reach the maximum.

We can decompose $f-g = (f-g)^+ - (f-g)^-$ where $h^+ = \max(0,h), h^- = -\min(h,0)$. So $\int (f-g)^+ = \int (f-g)^-$.

f, g is monotone means $f-g$ has bounded variation. In particular,

$$V(f-g) \le V(f) + V(g) \le 2.$$

So $\sup(f-g)^+ + \sup(f-g)^- \le 1$. To see this we can assume $f(0) = g(0) = 0, f(1) = g(1) = 1$ since we don't assume f, g to be continuous. And for simplicity assume supermum can be taken, say $f(a) - g(a) = \max(f-g)$, $f(b) - g(b) = \min(f-g)$.

Assume $a < b$ otherwise swap f, g. Then

$$\begin{aligned} V(f-g) &= V_0^a(f-g) + V_a^b(f-g) + V_b^1(f-g) \\ &\ge (\max - 0) + (\max - \min) + (0 - \min) \\ &= 2(\max - \min) \\ &= 2(\sup(f-g)^+ + \sup(f-g)^-). \end{aligned}$$

Then use

$$\int |f-g| = \int_{(f>g)} (f-g)^+ + \int_{(f<g)} (f-g)^- = 2\int_{(f>g)} (f-g)^+ =: 2I.$$

And we have

$$I \le m(f>g) \cdot \sup(f-g) \text{ and } I \le m(f<g) \cdot \sup(g-f)$$

with $m(f>g) + m(f<g) \le 1$, $\sup(f-g) + \sup(g-f) \le 1$.

If

$$I > \frac{1}{4},\ m(f>g)\sup(f-g) \ge I,$$

then

$$\begin{aligned} m(f<g)\sup(g-f) &\le (1 - m(f>g))(1 - \max(f-g)) \\ &\le 1 + m(f>g)\sup(f-g) - (m(f>g) + \sup(f-g)) \\ &\le 1 + I - 2\sqrt{I} = (1-\sqrt{I})^2 < I, \end{aligned}$$

contradiction. □

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